

Recall from the last few classes:

- We define the **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, denoted by $\det(A)$, by
$$\det(A) = ad - bc.$$
- **Theorem:** If A is an n by n invertible matrix, then for each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution; namely $\vec{x} = A^{-1}\vec{b}$.
- **Theorem:** Suppose A and B are n by n invertible matrices, then the following hold.
 1. A^{-1} is invertible and $(A^{-1})^{-1} = A$.
 2. $(AB)^{-1} = B^{-1}A^{-1}$.
 3. $(A^T)^{-1} = (A^{-1})^T$.
- An **elementary matrix** E is a matrix obtained from the identity matrix by a single elementary row operation.
- **Theorem:** A is invertible if and only if A is row equivalent to the identity matrix. Furthermore, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

1.

(a) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$. Find A^{-1} if possible.

(b) Let $A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$. Find A^{-1} if possible.

Invertible Matrix Theorem Let A be an $n \times n$ matrix, then the following are all equivalent.

- (a) A is invertible.
- (b) A is row-equivalent to I_n .
- (c) A has n pivot positions.
- (d) $A\vec{x} = \vec{0}$ has only the trivial solution.
- (e) The columns of A are linearly independent.
- (f) $x \mapsto A\vec{x}$ is one-to-one (injective).
- (g) For all $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ is consistent.
- (h) The columns of A span $\mathbb{R}^n$.
- (i) $\vec{x} \mapsto A\vec{x}$ is onto (surjective).
- (j) There exists a matrix C such that $CA = I_n$.
- (k) There exists a matrix D such that $AD = I_n$.
- (l) A^T is invertible.

2. Prove the theorem.

3. Determine if the following matrices are invertible.

(a) Let $A = \begin{bmatrix} 3 & 0 & 0 \\ -3 & -4 & 0 \\ 8 & 5 & -3 \end{bmatrix}$.

(b) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Theorem: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be linear and let A be the associated matrix. Then T is invertible if and only if A is invertible.

4. Prove the theorem.

Goal: Develop a numerical test for invertibility.

5. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Find a condition on the entries of A for A to be invertible.

Definition 1. For a matrix A define A_{ij} to be the matrix obtained from A by deleting row i and column j .

Remark:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a \det A_{11} - b \det A_{12} + c \det A_{13}.$$

Definition 2. For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{n+1} a_{1n} \det(A_{1n}) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}).$$

6. Compute the determinant of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Definition 3. The (i, j) -**cofactor** of A is given by $c_{ij} = (-1)^{i+j} \det A_{ij}$, and the **cofactor expansion along row i** is given by $\sum_{j=1}^n a_{ij} c_{ij}$.

Theorem 3.1: Cofactor expansion across any row or column is the same, and hence each cofactor expansion gives the determinant.

7. Compute the determinant of $C = \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix}$.

Definition 4. Let $A = [a_{ij}]$ be an $n \times n$ matrix, then A is **upper triangular** if $a_{ij} = 0$ when $i > j$.

Theorem 3.2: Let A be an $n \times n$ upper triangular matrix, then

$$\det(A) = a_{11}a_{22} \dots a_{nn} = \prod_{j=1}^n a_{jj}.$$

8. Prove the theorem.

Determinants – Answers / Solutions

1. (a) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$. We try augment A by I_n and try to use the result that if A is invertible, then $[A|I_n] \sim [I_n|A^{-1}]$. Thus

$$\left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ -1 & 5 & 6 & 0 & 1 & 0 \\ 5 & -4 & 5 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 6 & 10 & -5 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 0 & 0 & -7 & -2 & 1 \end{array} \right),$$

and now we're stuck. Since A is not row-equivalent to I_n (which we're seeing on left), the algorithm fails to find an inverse for A . Thus we know A is not invertible.

- (b) Let $A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$. We again augment A by I_n and try to use the result that if A is invertible, then $[A|I_n] \sim [I_n|A^{-1}]$. Thus

$$\begin{aligned} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array} \right) &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{array} \right) \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array} \right) \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{array} \right), \end{aligned}$$

and hence $A^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$.

2. *Proof.* There are many ways to organize all the implications. Here I try to use previous results from class as much as possible.

(a) $\implies$ (j) by definition.

(j) $\implies$ (d) by Theorem 2.5 (or above on this handout).

(d) $\implies$ (c) by Theorem 1.4.

(c) $\implies$ (b) since A is square.

(b) $\implies$ (a) by Theorem 2.7 (or above on this handout).

Thus we have established (a) $\iff$ (b) $\iff$ (c) $\iff$ (d) $\iff$ (j).

Note (a) $\implies$ (k) by definition of invertibility. We prove (k) $\implies$ (g). Suppose D is an $n \times n$ matrix such that $AD = I_n$. Let $\vec{b} \in \mathbb{R}^n$, and define $\vec{y} = D\vec{b}$. Now $A\vec{y} = AD\vec{b} = I_n\vec{b} = \vec{b}$, and hence $\vec{y}$ is a solution to the matrix equation $A\vec{x} = \vec{b}$. Thus the matrix equation is always

consistent, and hence (g) follows. Now we prove (g) $\implies$ (a). By Theorem 1.4, $A\vec{x} = \vec{b}$ being consistent for any $\vec{b}$ is equivalent to A having a pivot position in each row. Since A is square, we know A must be row equivalent to the identity matrix. By the theorem above, A is invertible, and hence (a) holds.

Finally note (d) $\iff$ (e) $\iff$ (f) and (g) $\iff$ (h) $\iff$ (i) follows from Theorem 1.12. Finally we prove (a) $\iff$ (l) above on this handout. $\square$

3. (a) We could row-reduce A to determine if A is row equivalent to the identity matrix, but in this case we can see the columns of A are linearly independent, and hence we know A must be invertible from the invertible matrix theorem.
- (b) We have previously seen this matrix does not span all of $\mathbb{R}^3$, and hence we can use the invertible matrix theorem to conclude A is not invertible.
4. *Proof.* We prove both implications directly from the definitions.

$\implies$ Suppose T is invertible, then there exists a function $T^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $T(T^{-1}(\vec{x})) = \vec{x}$ and $T^{-1}(T(\vec{x})) = \vec{x}$ for all $\vec{x} \in \mathbb{R}^n$. If we knew that T^{-1} was linear, then it would have an associated matrix we could use.

Claim: T^{-1} is linear.

Proof of Claim: Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. Since T is invertible, it is bijective. Thus there exists unique $\vec{a}, \vec{b} \in \mathbb{R}^n$ such that $T(\vec{a}) = \vec{x}$ and $T(\vec{b}) = \vec{y}$. Now we check the conditions for linearity:

$$T^{-1}(\vec{x} + \vec{y}) = T^{-1}(T(\vec{a}) + T(\vec{b})) = T^{-1}(T(\vec{a} + \vec{b})) = \vec{a} + \vec{b} = T^{-1}(\vec{x}) + T^{-1}(\vec{y}).$$

and

$$T^{-1}(c\vec{x}) = T^{-1}(cT(\vec{a})) = T^{-1}(T(c\vec{a})) = c\vec{a} = cT^{-1}(\vec{x}).$$

Thus T^{-1} is linear.

Since T^{-1} is linear, it has an associated matrix D . Now

$$AD\vec{x} = T(T^{-1}(\vec{x})) = \vec{x}$$

and

$$DA\vec{x} = T^{-1}(T(\vec{x})) = \vec{x}.$$

Thus D is the inverse matrix of A , and hence A is invertible.

$\impliedby$ Suppose A is invertible. Define $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $S(\vec{x}) = A^{-1}\vec{x}$. Then $T(S(\vec{x})) = AA^{-1}\vec{x} = \vec{x}$ and $S(T(\vec{x})) = A^{-1}A\vec{x} = \vec{x}$. Thus S is the inverse function of T , and hence T is invertible. $\square$

5. By the invertible matrix theorem, we know A is invertible if and only if A is row equivalent to the identity matrix. Thus we try to row-reduce A and see what condition we get on the entries of the matrix.

If the first column of A is all zeros, then we know the matrix is not invertible. Thus we assume $a \neq 0$.

$$\begin{aligned}
A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} &\sim \begin{bmatrix} a & b & c \\ ad & ae & af \\ ag & ah & ai \end{bmatrix} \sim \begin{bmatrix} a & b & c \\ 0 & ae - bd & af - cd \\ 0 & ah - bg & ai - cg \end{bmatrix} \\
&\sim \begin{bmatrix} a & b & c \\ 0 & ae - bd & af - cd \\ 0 & (ah - bg)(ae - bd) & (ai - cg)(ae - bd) \end{bmatrix} \\
&\sim \begin{bmatrix} a & b & c \\ 0 & ae - bd & af - cd \\ 0 & 0 & D \end{bmatrix}
\end{aligned}$$

where $D = (ai - cg)(ae - bd) - (af - cd)(ah - bg)$.

Now multiplying out D and simplifying we get

$$D = a(aei - ceg - bdi - afh + cdh + bfg).$$

Since a is non-zero, we know for A to be row equivalent to the identity matrix, we need $(aei - acg - bdi - afh + cdh + bfg) \neq 0$. We define

$$\det(A) = (aei - acg - bdi - afh + cdh + bfg).$$

This seems like a strange condition, so let's play around with it a bit more. Factoring we get the following

$$(aei - acg - bdi - afh + cdh + bfg) = a(ei - fh) - b(di - fg) + c(dh - eg),$$

and now we see some smaller determinants showing up; namely

$$a(ei - fh) - b(di - fg) + c(dh - eg) = a \cdot \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \cdot \det \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \cdot \det \begin{bmatrix} d & e \\ g & h \end{bmatrix} = \det(A).$$

This expression for the determinants suggests a way to define the determinant of an $n \times n$ matrix in terms of $(n - 1) \times (n - 1)$ matrices.

6. We use the definition to compute the determinant:

$$\begin{aligned}
\det A &= \det \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = 1 \cdot \det \begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix} - 2 \cdot \det \begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix} + 3 \cdot \det \begin{bmatrix} 4 & 5 \\ 7 & 8 \end{bmatrix} \\
&= 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = 0.
\end{aligned}$$

We have previously seen that A is not invertible, so it is reassuring to see the determinant is also zero.

7. Again we use the definition to compute the determinant:

$$\det C = \det \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix} = 6 \cdot \det \begin{bmatrix} 7 & 2 & -5 \\ 0 & 0 & 0 \\ 3 & 1 & 8 \end{bmatrix} - 5 \cdot \det \begin{bmatrix} 1 & 7 & 2 \\ 2 & 0 & 0 \\ 8 & 3 & 1 \end{bmatrix} = -5(-2) = 10.$$

Using the last Theorem, we can expand about the third row as well. We show that here:

$$\det C = \det \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix} = 2 \cdot \det \begin{bmatrix} 0 & 0 & 5 \\ 7 & 2 & -5 \\ 3 & 1 & 8 \end{bmatrix} = 2(5)(7 - 6) = 10.$$

8. *Proof.* We prove the result by induction on the size of the matrix.

Base Case: Let $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$, then $\det(A) = ac$. Thus the result holds for 2×2 matrices.

Inductive Step: We assume that the determinant of an $n \times n$ upper triangular matrix is given by the product of the diagonal elements.

Let A be an $(n + 1) \times (n + 1)$ upper triangular matrix. We compute the determinant by expanding along the first column:

$$\det(A) = a_{11} \det(A_{11}) - a_{21} \det(A_{21}) + \cdots + (-1)^{n+1+1} a_{(n+1)1} \det(A_{(n+1)1})$$

Since A is upper-triangular, $a_{21} = \cdots = a_{(n+1)1} = 0$. Thus

$$\det(A) = a_{11} \det(A_{11}).$$

Since A_{11} is an $n \times n$ upper triangular matrix, we can apply the induction hypothesis to get

$$\det(A) = a_{11} \det(A_{11}) = a_{11}(a_{22} \cdots a_{(n+1)(n+1)}).$$

Thus the result holds by mathematical induction. □